

|  |           |                                                                                                                                                       |
|--|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |           | When $n$ and $T$ are decreased by half, gradient increases by 4.                                                                                      |
|  | <b>13</b> | <b>D</b> $Q = (mc\Delta\theta)_{ice} + ml_f + (mc\Delta\theta)_{water}$ $= 2.1(2108)(4.5) + 2.1(334 \times 10^3) + 2.1(4190)(96)$ $= 1.57 \text{ MJ}$ |
|  | <b>14</b> | <b>C</b> The volume of the molecules is much smaller in comparison to the volume of container.                                                        |